

Atherv Vidhate

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Education

University of California, San Diego

San Diego, CA

Bachelor of Science in Data Science, Minor in Business Analytics

Expected Graduation: 2027

- Tau Kappa Epsilon Historian: Built alumni engagement portal to support donations, networking, and access to chapter history for 1000+ members; implemented SDLC practices, conventional commits, and code review standards.
- Economics Lab Research Assistant: engineered a data pipeline using the Library of Congress API to extract 100,000+ entries on 19th-century presidential campaign newspaper coverage and group 300+ related publications.

Experience

Machine Learning Engineer Intern

June 2026 – Sep 2026

Keysight Technologies

- Developed a product-quarter demand anomaly detection pipeline combining Isolation Forest with a 7+ model PyOD ensemble establishing multi-model agreement scoring to reduce false-positive anomaly flags.
- Built deep learning anomaly detectors, including a tabular autoencoder and quarterly LSTM autoencoder over engineered sequence-window datasets, to benchmark unsupervised anomaly detection against traditional ML baselines.
- Engineered customer concentration and pipeline-context explainability features, enabling anomalies to be classified as customer-driven, broad-based, or pipeline-supported for business stakeholders.

AI Research & Development Intern

Jan 2026 – June 2026

Intellinum

- Architected a 6-agent autonomous AI pipeline using Redis Streams and the Claude API to automate enterprise page generation and Oracle Fusion SCM integration, eliminating manual development cycles across 16+ REST API endpoints.
- Engineered modular LLM agent prompt systems with structured tool-file decomposition, shared identity contracts, and a 3-step grounding protocol, enabling agents to produce zero-defect production JSON artifacts with no human intervention.
- Developed a rules-based validation engine that statically analyzes FlexiPro Studio page JSON, detecting and triaging issues across 3 severity levels with structured output routed to the appropriate engineering, QA, or integration teams.

Full-Stack Developer Intern

Aug 2025 – Jan 2026

Nexa Tax

- Delivered 95% structured data extraction accuracy in production-grade AI tax processing system with 100+ PDF documents processed through RAG pipeline, achieving structured data extraction from financial documents with 23 categories.
- Optimized computer vision pipeline achieving 60% reduction in processing time through YOLO object detection and multimodal LLM architecture for receipt/income document processing, with 100% uptime for 1000+ concurrent users.
- Engineered scalable Flask REST API with LangChain and LangGraph state management, implementing 8+ endpoints for chatbot interaction and conversation history tracking with Postgres persistence and user session management.

Projects

Fitness Program Recommendation System | workout.atherv.com

- Built a recommendation engine over 600,000+ rows with K-Means clustering, cosine similarity, and PCA for visualization.
- Fine-tuned 2 BERT models and engineered a custom reusable embedding module that outputs sentence-level vectors.
- Shipped with Docker and Google Cloud Platform for deployment and automated CI/CD pipeline with Github Actions.

Concert Scout AI | concertscout.app

- Built a full-stack AI-powered concert recommendation system using FastAPI and Next.js for over 300 active users.
- Implemented a multi-agent AI pipeline with Google Gemini to deliver personalized concert results.
- Deployed a scalable backend infrastructure with Redis session management and analytics tracking.

Power Outage Analysis | atherv.com/poweroutageanalysis

- Used techniques like one-hot encoding, hyperparameter tuning, and cross-validation to build optimized Random Forest models predicting outage severity, achieving a 15% accuracy improvement.
- Cleaned 1,500+ row dataset, ensuring structured formats and engineering 4 new features for enhanced analysis.

Technical Skills

Programming Languages: Python, SQL, R, Java, TypeScript, JavaScript, HTML/CSS

Frameworks & Libraries: scikit-learn, PyTorch, TensorFlow, Spark, LangChain, pandas, Matplotlib, NumPy, Dask, FastAPI

Tools & Platforms: Git, Docker, AWS, GCP, Redis, PostgreSQL, Tableau, Power BI, Excel